

HumanEval

Beginning

Mean depth 3.31

```
<think>
Okay, I need to solve this programming problem. Let's read the task carefully. The function x_or_y takes three parameters: n, x, and y. The goal is to return x if n is a prime number, otherwise return y.
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```
So the first thing I need to do is figure out whether the given n is a prime. Then, based on that, return the appropriate value.
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```
Wait, but what about the value of n? For example, if n is 1, which is not a prime. So the function should
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Ending

Mean depth 1.04

```
"""A simple program which should return the value of x if n is
a prime number and should return the value of y otherwise.
```

```
Examples:
for x_or_y(7, 34, 12) == 34
for x_or_y(15, 8, 5) == 5
```

```
"""
def is_prime(n):
    if n < 2:
        return False
    if n == 2:
        return True
    if n % 2 == 0:
        return False
    for i in range(3, int(n**0.5) + 1, 2):
        if n % i == 0:
            return False
    return True
```

```
if is_prime(n):
    return x
else:
    return y
...
```

Iteration depth

1 2 3 4 5 6 7 8

OlympiadBench

Beginning

Mean depth 2.46

```
<think>
Okay, so I need to figure out the value of f(2,5) given the function f(a, b) = a/b + b/a + 1/(ab). Let me start by understanding what the function does. It takes two positive integers a and b and adds three terms: the first term is a divided by b, the second is b divided by a, and the third is 1 divided by the product of a and b.
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```
Let me write down the function again to make sure I have it right: f(a, b) = (a/b) + (b/a) + (1/(ab)). Yeah, that's what it says. So for f(2,5), I just
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Ending

Mean depth 1.29

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Combine the first two terms:
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$$
\frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab}
$$
```

```
Then add the third term:
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$$
f(a, b) = \frac{a^2 + b^2}{ab} + \frac{1}{ab} = \frac{a^2 + b^2 + 1}{ab}
$$
```

```
Apply this simplified form to $ a = 2 $, $ b = 5 $:
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$$
f(2, 5) = \frac{2^2 + 5^2 + 1}{2 \cdot 5} = \frac{4 + 25 + 1}{10} = \frac{30}{10} = 3
$$
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```

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### Final Answer
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$$
\boxed{3}
$$
```

GPQA

Beginning

Mean depth 3.47

```
<think>
Okay, so I need to find the eigenvalues of the Hamiltonian operator H, which is given by H = ε · σ · n. Let me recall what I know about Pauli matrices and eigenvalues.
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First, the Pauli matrices σ are 2x2 complex matrices. They have eigenvalues ±1, right? Each of the σ matrices has eigenvalues 1 and -1. But here, the Hamiltonian is a scalar multiple of the vector product of ε (which is an energy constant) and the unit vector n. So, H is ε times the vector n dotted
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...

Ending

Mean depth 3.69

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a scalar, and multiplying it by ε gives a scalar, which is equivalent to a scalar multiple of the identity matrix.
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### Eigenvalues of the Hamiltonian
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Since H is a scalar multiple of the identity matrix, it is a **diagonal matrix** in the standard basis. Therefore, its eigenvalues are simply the scalar multiples of the eigenvalues of the identity matrix, which are 1 and -1.
```

```
Thus, the eigenvalues of H are:
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$$
\pm \epsilon \cdot |\mathbf{n}|
$$
```

```
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```

```
### Final Answer
```

```
$$
\boxed{A}
$$
```